Dr. Thejus Mahajan
Reventlowstraße 17
22605 Hamburg
+49-15259697629
thejus.mahajan@uni-hamburg.de

Hamburg, 10 March 2026

Frau Dr. Pooja Gupta
Department of Stem Cell Biology
Universitätsklinikum Erlangen
Kussmaulallee 4
91054 Erlangen

Postdoctoral Researcher – AI-based Multi-Omics in Computational Biomedicine

Dear Dr. Gupta,

Three years of experience translating continuous biological models into high-performance Python architectures have prepared me to build the AI frameworks required for your BMFTR-funded project, “AI-Driven Multi-Omics Integration for Predicting IBD-PD Comorbidity Progression”. Your group’s focus on utilizing machine learning to decode the gut-brain axis and comorbidity networks targets exactly the intersection of complex high-dimensional systems and deep learning that I am passionate about. My transition from computational systems modeling to bioinformatics pipeline engineering makes me a highly suitable candidate to build the AI frameworks your research requires.

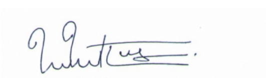
While my academic background is in systems modeling, I have spent the last year intensively retraining in modern bioinformatics and machine learning. At my current internship with HealthTwiSt GmbH, I am solely responsible for refactoring a production medical data pipeline (in R/tidyverse) for Germany’s interventional radiology quality registry. Working with 143,000 patient records and 1,920 variables, I externalized hardcoded logic into configuration files, verified byte-level reproducibility, and reduced the 1,834-line codebase by 26%, dramatically improving maintainability.

Your project’s need for graph neural networks, variational autoencoders, and integrative molecular interaction networks aligns closely with my extensive experience analyzing coupled systems. I bring a strong foundation in the tools you require:

- **AI / Deep Learning Foundations:** During my post-doc, I led the translation of a legacy Fortran model into a high-performance Python engine using **Google JAX**. By engineering branchless logic and vectorized multidimensional array operations, I achieved direct compatibility with **Google TPUs and GPUs** for massive parallelization. As JAX shares core paradigms with **PyTorch** and **TensorFlow**, this provided the foundational expertise required to scale the graph neural networks and autoencoders your project demands.
- **Pipeline Development & ML Workflows:** Actively building ML workflows for tabular clinical data using Tidymodels (R) and expanding my deep learning capabilities via IBM’s “Machine Learning with Python” and “Introduction to AI” training. Highly proficient in building reproducible workflows (Nextflow) and processing multi-omic data.
- **Large-Scale Data & High-Performance Computing:** Processed terabytes of raw experimental data during my Ph.D. and developed algorithms for HPC clusters. My expertise navigating complex, integrated networks and running simulation-based perturbation analyses aligns perfectly with modeling IBD-PD comorbidity progression.

I am deeply motivated by the Universitätsklinikum Erlangen’s mission to leverage AI for biomarker discovery and translational impact in complex diseases. My robust mathematical background combined with modern data engineering skills will allow me to immediately contribute to your computational infrastructure. I welcome the opportunity to discuss further and am available for an interview.

Sincerely



Dr. Thejus Mahajan

THEJUS MAHAJAN PH.D.

Bioinformatics & Data Scientist

+49-15259697629
thejus.mahajan@uni-hamburg.de
thejusmahajan.github.io
Hamburg, Germany

*30.09.1992 (Talikulam, India)
Married

PROFILE

- **Focus:** Large-scale data analysis, bioinformatics pipeline development, and statistical modeling.
- **Core Stack:** R (Bioconductor, Tidymodels), Python (Pandas, JAX, Scikit-learn), Nextflow (DSL2), BASH.
- **Current Objective:** Applying computational and machine learning expertise in single-cell multi-omics to advance research in stem cell biology.

EXPERIENCE

Internship – Bioinformatics & Data Pipeline Engineering
HealthTwiSt GmbH
02/2026 - 04/2026 Berlin, Germany

Clinical data research and development under Dr. Andreas Busjahn
- Refactored a 1,834-line medical data pipeline (R/tidyverse) for Germany's interventional radiology quality registry (DeGIR); 143K rows $\times$ 1,920 columns.
- Externalized 257 hardcoded rules into 8 config files; 26.5% code reduction with byte-level output verification (`identical()`).
- Identified 2 latent bugs through systematic code analysis; documented without altering pipeline behavior.
- Machine Learning workflows using the Tidymodels framework in R.

Further Training – Bioinformatics and Biostatistics
CQ Beratung + Bildung GmbH
08/2025 - 02/2026 Berlin, Germany

Application-related Bioinformatics and Biostatistics
- Programming methodology: Shell (BASH), Python (object-oriented), R.
- Bioinformatics databases (NCBI, Biopython, SQL) and sequence analysis.
- NGS analysis pipelines (Nextflow, Galaxy); structural bioinformatics.
- Clinical biostatistics (R/Bioconductor, ANOVA, PCA, HCA, survival analysis); GDPR/BDSG compliance.

Guest Scientist
Helmholtz-Zentrum Hereon, Ökosystemmodellierung
05/2025 - 10/2025 Geesthacht, Germany

Computational Biologist & Systems Modeler
- Spearheaded research modeling evolutionary processes in marine ecosystems.
- Applied advanced computational techniques to simulate long-term environmental impacts.

Job Search & German Language Learning
Hamburg, Germany
02/2025 - 04/2025 Hamburg, Germany

Active job search and German language acquisition (Goethe B1 certification).

Post-doctoral Scientist
University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences
08/2021 - 01/2025 Hamburg, Germany

Marine Ecosystem Modelling
- *Elternzeit (Parental Leave) from 01/2024 to 12/2024.*

STRENGTHS

Data Pipeline Engineering
Refactored a production medical data pipeline (1,834 lines, 143K rows), achieving 26.5% code reduction with zero behavioral regression. Byte-level verification after every change.

Complex Problem-Solving
Resolved a critical data-loss error by creating a Fortran simulation to correct for a misplaced particle detector, recalculating results using multi-sensor data.

Bioinformatics & Biostatistics
Bridged physical science expertise with modern biological workflows. Skilled in complex multidimensional array operations, JAX/GPU acceleration, sequence alignment, and multivariate statistical modeling using R (Bioconductor) and Python.

TECHNICAL SKILLS

Programming
Python (JAX, Pandas, NumPy, Scikit-learn, Biopython), R (Bioconductor, Tidymodels), SQL, BASH

Bioinformatics
Nextflow (DSL2), NGS Analysis, Multiple Sequence Alignment (MAFFT), Sequence Trimming (trimAl), NCBI / Entrez

Biostatistics & ML
Deep Learning Foundations (TensorFlow, PyTorch concepts via JAX), Multivariate Analysis, PCA, Clustering, Scikit-learn, Tidymodels

Data Engineering
Hardware Acceleration (TPU/GPU), Large dataset processing (HDF5, NetCDF), Pipeline development

Tools
Linux/Unix, Git, HPC clusters, Conda

TRAINING / COURSES

IBM: Machine Learning with Python; Introduction to AI
Coursera (in progress)

High Performance Computing (HPC) Training
Jülich Supercomputing **Center (JSC)**,
Forschungszentrum Jülich

- Model Development: Developed the novel "Cyanobacteria Life Cycle" (CLC) model within the ERGOM framework for climate warming projections.
- High-Performance Computing & ML: Led the translation of a legacy Fortran continuous model into a high-performance Python simulation engine. Utilized **Google JAX** vectorization and branchless logic for massive parallelization (direct compatibility with **TPUs and GPUs**).
- This architecture established a strong foundation in tensor operations underpinning deep learning frameworks like **PyTorch** and **TensorFlow**.

Job Search
Kerala, India
04/2021 - 07/2021 · Kerala, India

Relocation from India to Germany and active job search.

Physics Educator
Tutorwaves Solutions Inc / Freelance
10/2018 - 03/2021 · Kerala, India

Prepared physics content for competitive examinations and coached state-level chess teams.

Ph.D. Researcher in Astrochemistry
University of Paris-Saclay
10/2015 - 09/2018 · Orsay, France

Led atomic collision experiments, analyzed data, and modeled outcomes.
- Acquired and processed terabytes of raw data from tandem particle accelerator experiments.
- C++ signal processing, algorithm development, and molecular fragmentation modeling (KIDA database).

EDUCATION

Ph.D. in Astrochemistry
Université Paris-Saclay, Institute of Molecular Sciences (ISMO) Orsay, France
09/2015 - 09/2018 · Orsay, France

M.Sc. in Physics
National Institute of Technology Calicut, India
07/2012 - 01/2015 · Calicut, India

B.Sc. in Physics
University of Calicut, India
06/2009 - 04/2012 · Thrissur, India

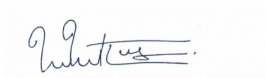
REFERENCES

Dr. Andreas Busjahn
abusjahn@healthtwist.de

Prof. Dr. Kai Wirtz
kai.wirtz@hereon.de

Prof. Dr. Elisa Schaum
elisa.schaum@uni-hamburg.de

Hamburg, 10 March 2026



Dr. Thejus Mahajan

MITx: 7.00x Introduction to Biology - The Secret of Life
Comprehensive coursework on genetics, biochemistry, and molecular biology.

OTHER SELECTED PROJECTS

Hepatitis Delta Virus Pipeline • GitHub
Nextflow (DSL2) pipeline for viral sequence analysis: NCBI Entrez download, MAFFT alignment, trimAl trimming, Conda-managed dependencies.

JAX/TPU High-Performance Simulator • GitHub
Ported a legacy Fortran model to a modular Python architecture using Google JAX. Engineered branchless, vectorized logic for massive TPU/GPU parallelization.

Introduction to Tidymodels • GitHub
Teaching script for ML in R: recipes, parsnip, workflows, and yardstick on the Palmer Penguins dataset.

Berlin Tree Analysis • GitHub
Interactive Jupyter notebook analyzing 5 years of tree planting data in Berlin using Python, Pandas, and Seaborn.

PUBLICATIONS

5 peer-reviewed articles • Google Scholar Profile

LANGUAGES

English
Proficient

German
B1 - Goethe Certified

French
Intermediate

Malayalam
Native

Tamil
Proficient

Hindi
Intermediate